

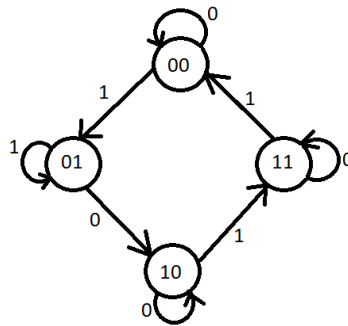
Asm4

Submission:

<https://docs.google.com/forms/d/e/1FAIpQLSfQ-Mcb4H-WK-m8ZrOgitlmKquvreBFNHIBM3MqXEOfQjgZug/viewform?usp=header>

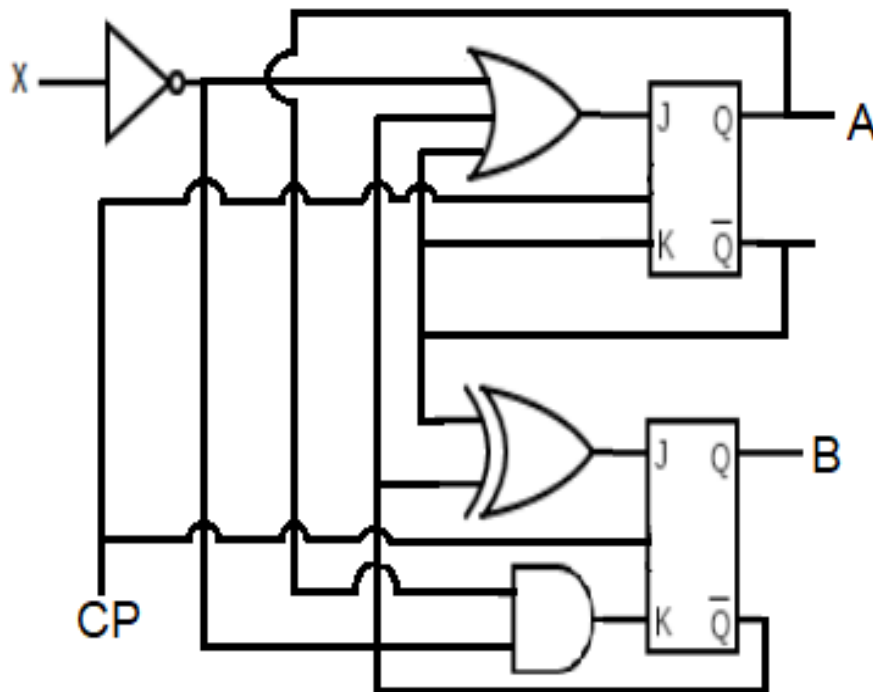
Assignment 4:

- 1) Given the state diagram as follows, get the sequential circuit using D flipflop.



**Assignment 5 : Extra[Only If anyone missed any prev asms]**

- 1) Draw the state diagram for the given circuit.



Assm3

## Assignment 3

1. Design a circuit diagram for the following system that takes two 4-bit binary numbers  $A$  and  $B$  as inputs and outputs in the following fashion:
  - If  $A$  is divisible by 4 and  $B$  is greater than 7 then the output should be  $A-B$
  - $A+B$  otherwise
2. Design a circuit diagram for the following system that takes two 4-bit binary numbers  $A$  and  $B$  as inputs and outputs in the following fashion. Suppose all the gates are 2 input gates. Minimize the number of gates in your circuit.
  - If both numbers are divisible by 3 the output should be  $A+B$
  - $A-B$  otherwise

Submission Link:

<https://docs.google.com/forms/d/e/1FAIpQLSf0-hrP0D4g0J9RRAKl7spVPiL9iDALLThEqm-FqQrA148etA/viewform?usp=header>

Try them. Submit whatever you tried. Submit in time.

Asm2

### Question 1

On planet Z, there have been two new discoveries of Condition A and Condition B in plants. You have been tasked with creating a system which will detect the chances of development of Conditions A and B in plants. The conditions depend on the following factors:

Inputs (Factors) :

W: High Temperature. (W=1 if the temperature is high, W=0 otherwise)

X: Oxygen Level (X=1 if oxygen level is normal, X=0 otherwise)

Y: Presence of UV Light (Y=1 if UV light is present, Y=0 otherwise)

Z :Humidity (Z= 1 if the humidity is high, Z=0 otherwise)

Outputs: A : Condition

B : Condition

1. A plant will develop Condition A if **presence of UV Light** is accompanied by
  - i) **a high temperature with normal oxygen level**
  - or
  - ii) **low humidity.**
2. If **UV Light is absent** then the plant will develop Condition A if there is
  - i) **an abnormal level of oxygen**
  - or
  - ii) **there is high humidity with low temperature**
3. If the **oxygen level is not normal** or **there is high humidity with low temperature** then the plant will develop Condition B.

Devise a truth table for the system above which will detect the presence of the conditions based on the given factors. In your truth table, the input columns should be in the order WXYZ, where W is the MSB and Z is the LSB.

Draw K-map (s) based on the scenario above and derive simplified output (s).  
Also draw the circuit diagram.